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2000 Recert

ABT RECERTIFICATION EXAMINATION #17

May 12, 2000 ***PART A***

This examination constitutes part of the recertification process for toxicologists certified by the American Board of Toxicology in 1991 and 1996. There are three parts in this exam and an answer sheet is provided for each part. These answer sheets must be returned to ABT, P.O. Box 30054, Raleigh, NC 27622 for grading. An analysis of your performance in each part of the exam will be provided to the recertification committee of ABT and they will contact you regarding this and other requirements of the recertification process. If you have any questions please call the ABT Office at 919-847-8720.

EXAMINATION CATEGORIES

1. Metals, solvents, pesticides, gases, dusts and aerosols, industrial chemicals and radiation, plant and animal toxins, food additives, drugs and cosmetics.
2. Mutagenesis, carcinogenesis, developmental and reproductive toxicology, pulmonary toxicology, neurologic toxicology, immunologic toxicology, cutaneous toxicology, ocular toxicology, hepatic toxicology, hematopoietic toxicology, renal toxicology.
3. General principles, toxicokinetics, toxicodynamics and mechanisms of action, risk assessment, epidemiology and biostatistics, regulatory toxicology and risk management, environmental toxicology, occupational toxicology, forensic and clinical toxicology.

INSTRUCTIONS

1. Use a soft (#2) pencil to fill the box corresponding to your answer.
2. Select only one answer for each question. (Erase others)
3. Do not leave any questions unanswered. (No penalty for guessing).
4. Sign answer sheet ONLY on line that says name.
5. Do not bend answer sheets.

Good luck!

Key for the following questions
Choose the single most appropriate answer

1. Which of the following pairs of radionuclides are of concern because they are metabolic analogs of calcium, deposited in bone, and may cause skeletal neoplasia?
   A) ^{222}Rn and ^{137}Cs
   B) ^{222}Rn and ^{90}Sr
   C) ^{226}Ra and ^{90}Sr
   D) ^{137}Cs and ^{90}Sr
   E) ^{238}U and ^{238}Pu

2. A semi-conscious pesticide sprayer is brought into the emergency room from a nearby orchard with acute carbaryl poisoning that occurred within the last 45 minutes. He is starting to have trouble breathing. In addition to putting him on a mechanical respirator and cleansing his skin of any remaining pesticide residues, which of the following treatments is indicated?
   A) propanolol followed by an intravenous anti-inflamatory steroid as needed
   B) 2-PAM (pralidoximine) followed by atropine as needed
   C) atropine as needed, followed by 2-PAM
   D) atropine only, repeated as needed
   E) supportive therapy only, possibly with theophylline to keep the airways open

3. Botulinum toxin inhibits neuromuscular communication by
   A) blocking sodium channels along the motor nerve
   B) inhibiting acetylcholine release from the axon terminal
   C) inhibiting the reuptake of choline resulting in acetylcholine depletion
   D) competitively blocking postsynaptic acetylcholine receptors
   E) evoking unstimulated acetylcholine release

4. Some elements can be methylated by metabolic pathways in vertebrate species, or by microbes in water, sediment or soils. Important products of biomethylation include
   A) trimethyl tin, tetraethyl lead, methyl mercury
   B) methyl mercury, dimethyl arsine, dimethyl selenide
   C) tetramethyl tin, methyl mercury, dimethyl arsine
   D) only methyl mercury
   E) methyl mercury, dimethyl cadmium, tetraethyl lead

5. Contact urticaria is characteristic of
   A) mistletoe
   B) crocus
   C) pokeweed
   D) nettles
   E) poison ivy

6. The dominant route of excretion for absorbed lead in adult humans is
   A) bile
   B) feces
   C) urine
   D) epithelial tissue
   E) respiratory tract

7. Bagassosis is a term for allergic pulmonary responses occuring as a result of occupational inhalation exposure to
   A) cotton dust
   B) coal dust
   C) beryllium
   D) hay spores
   E) sugar cane dust

8. During tidal breathing, inhalation of a monodispersed particle about 1 μm in diameter undergoes primary deposition in the
   A) bronchus
   B) trachea
   C) nasopharyngeal region
   D) alveolus
   E) pleura

9. Which of the following is NOT characteristic of a tricyclic antidepressant overdose?
   A) disorientation, ataxia, depression
   B) dilated pupils, hyperpyrexia and dry mouth
   C) decreased PR interval and decreased QRS interval on ECG
   D) cardiac arrhythmia with ventricular tachycardia
   E) urinary retention and decreased gastrointestinal motility

10. Ingestion of which of the following radioactive materials would be most damaging to the bone marrow?
   A) beta emitters
   B) alpha emitters
   C) gamma emitters
   D) those with highest energy regardless of type of emission
   E) elements of medium Z number since their radiations have the most favorable charge to mass ratio

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11. Which of the following chemicals is NOT an antioxidant?
A) vitamin E
B) ethoxyquin
C) riboflavin
D) butylated hydroxytoluene
E) propyl gallate

12. Intranuclear inclusions in the renal tubular epithelial cells is characteristic of
A) cobalt
B) mercury
C) antimony
D) lead
E) cadmium

13. In some species, accumulation of neurofibrillary tangles in the brain is associated with exposure to
A) lead
B) manganese
C) methyl mercury
D) aluminum
E) triethyltin

14. In the treatment of cyanide (CN) poisoning, nitrite acts by
A) displacing CN from cytochrome oxidase
B) oxidizing the iron in cytochome oxidase from Fe^{2+} to Fe^{3+}
C) oxidizing CN to thiocyanate
D) oxidizing the iron in hemoglobin from Fe^{2+} to Fe^{3+}
E) displacing the CN from cytochome P450

15. Which of the following classes of antibiotics is most commonly associated with ototoxicity and nephrotoxicity?
A) penicillins
B) fluoroquinolines
C) aminoglycosides
D) tetracyclines
E) macrolides

16. Elevated carboxyhemoglobin level occurs following exposure to
A) bis(chloromethyl)ether
B) methylene chloride
C) methyl ethyl ketone
D) methylene blue
E) methyl chloride

17. Chronic absorption of which of the following results in hyperostosis?
A) cyanide
B) sulfide
C) fluoride
D) borate
E) oxalate

18. The treatment of poisoning by organophosphates can be very difficult. One reason for this is the difficulty of reactivating the enzyme acetylcholinesterase because the organophosphates attach to the enzyme through
A) Van der Waals' bonds
B) ionic bonds
C) hydrogen bonds
D) covalent bonds
E) reinforced ionic bonds

19. The toxic glycoprotein in castor beans is
A) strychnine
B) alizarin
C) brucine
D) muscarine
E) ricin

20. Which of the following is a nonintentional food additive?
A) monosodium glutamate
B) diethyl phthalate
C) sodium benzoate
D) propylene glycol
E) sorbic acid

21. Loxoscelism can result from
A) eating improperly prepared salmon
B) a spider bite
C) eating cinchona bark
D) any teratogenic insult with concomitant torticollis
E) eating mushrooms

22. Which of the following is NOT true for carbon monoxide poisoning
A) signs and symptoms will not appear in a normal person exposed to 30 ppm regardless of the exposure time 0.003%
B) an individual with coronary vessel disease is more prone to develop angina than a normal individual with similar concentrations of carboxyhemoglobin
C) in the absence of coronary vessel disease carboxyhemoglobin at concentrations of less than 10% saturation will result in an increased coronary blood flow
D) the relapses occurring in persons recovering from acute CO poisoning are most likely due to CO storage in myoglobin
E) 20% carboxyhemglobin is associated with mild headache

23. The rate of oxidation of ethanol in the body
A) is proportional to blood ethanol concentration
B) can be accelerated significantly by the administration of oxygen
C) can be accelerated significantly by vigorous physical exercise
D) is equivalent to 10 ml ethanol per hour
E) all of the above

24. Which of the following is the first symptom of iron intoxication in humans?
A) bloody diarrhea
B) convulsions
C) acidosis and fever
D) hepatic necrosis
E) gastrointestinal obstruction

25. Xylene toxicity is characterized by
A) a relatively low acute systemic toxicity compared to other organic solvents
B) a high potential for chronic toxicity due to its effect on the bone marrow
C) bronchopneumonia subsequent to inhalation of vapor
D) an odor threshold exceeding its threshold limit value
E) all of the above

26. Severe necrosis may follow the bite of a
A) tarantula
B) brown recluse spider (*Loxosceles reclusa*)
C) black widow spider (*Latrodectus sp.*)
D) *Centruroides sp.*
E) tick

27. Which of the following is the major neurotoxic metabolite of n-hexane?
A) 2-hexanone
B) 2,5-hexanedione
C) 2-hexanol
D) 5-hydroxy-2-hexanone
E) 2,5-hexanediol

28. Which of the following nickel compounds is most toxic?
A) nickel oxide
B) nickel carbonyl
C) metallic nickel
D) nickel fluoride
E) nickel monosulphide

29. Mydriasis and cycloplegia are caused by the ingestion of plant alkaloids which contain
A) atropine
B) digitalis
C) beta aminopropionitrile
D) aconitine
E) pyrrolizidine alkaloids

30. A possible mechanism for excretion of amphetamine in the urine is
A) passive diffusion
B) ionization of a weak organic acid
C) enhancement by alkalization of the urine
D) ionization of a weak organic base
E) none of the above

31. The biological activity of actinomycin D is due to its
A) capacity to intercalate with DNA
B) damage to plasma membranes
C) incorporation into RNA
D) impairment of cellular transport
E) ability to cause mitochondrial dysfunction

32. Which of the following is true regarding food additives?
A) for regulatory purposes, food additives include only those substances added to foods for functional purposes
B) the Delaney ammendment of the Food, Drug Cosmetic Act prohibited the use of any food additive which has been shown to cause carcinogenic, mutagenic or teratogenic effects in laboratory animals
C) the FDA normally applies a 100-fold safety factor to the NOAEL obtained in appropriate animal toxicity tests
D) food additives are regulated by the Department of Agriculture under the Pure Food Act
E) even if there is no detectable migration of a packaging component into the food, the packaging component is still considered to be a food additive

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Key for the following questions
Choose A) if 1,2 and 3 are correct
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Choose C) if 2 and 4 are correct
Choose D) if only 4 is correct
Choose E) if all four answers are correct

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33. Characteristics that distinguish pyrethroid type II compounds from type I compounds are
1) mammalian toxicity is manifested by paresthesias
2) presence of a cyano group in the carboxylic acid moiety of the molecule
3) mammalian toxicity is manifested by writhing movement
4) difference in their relative potential to depolarize persistently and block conduction in nerves

34. Trivalent chromium
1) is an essential trace nutrient and cofactor for insulin action
2) is less toxic than hexavalent chromium
3) has adverse effects on renal tubules and glomerular apparatus at high concentrations
4) has been implicated along with hexavalent chromium in occupationally induced cancers of the respiratory system

35. Carbon disulfide
1) has been linked to coronary heart disease
2) is metabolized by cytochrome P450
3) produces peripheral neuropathy
4) is a degradation product of dithiocarbamate pesticides

36. Scombroid poisoning
1) is an anaphylactoid reaction
2) is effectively treated with antihistamines
3) is commonly caused by ingesting spoiled tuna
4) is an allergy to fish

37. 2,3,7,8-tetrachlorodibenzo-p-dioxin (TCDD)
1) has been found as a contaminant in 2,4,5-T
2) is extremely toxic to some animal species
3) is teratogenic and carcinogenic
4) induces microsomal enzymes

38. Which of the following toxins block(s) the inward flow of sodium through sodium channels along the axon?
1) botulinum toxin
2) tetrodotoxin
3) batrachotoxin
4) saxitoxin

39. Treatment of methanol poisoning may involve the use of the intravenous infusion of
1) isoproterenol
2) sodium bicarbonate
3) disulfiram
4) ethanol

40. Which of the following substances exerts its principal toxic effect on cholinergic transmission?
1) botulinum toxin
2) nicotine
3) atropine
4) ciguatoxin

41. Which of the following chemicals is/are formed from photochemical processes during episodes of air pollution?
1) ozone
2) carbon monoxide
3) peroxyacetyl nitrate
4) sulfur dioxide

42. Isopropanol poisoning should be treated by
1) inducing vomiting
2) administration of sodium bicarbonate
3) administration of ethanol
4) hemodialysis

43. Which of the following metals has/have carcinogenic activity?
1) lead
2) manganese
3) nickel
4) molybdenum

44. Aspirin may cause which of the following physiological and pathological responses?
1) respiratory alkalosis and compensatory metabolic acidosis
2) stimulation of cyclooxygenase
3) microvesicular fatty change of the liver
4) retention of sodium, potassium and water

45. The pulmonary pathology of chronic exposure to SO_2 consists of
1) loss of bronchoconstriction reflex
2) increase in mucus layer thickness
3) decrease in flow resistance
4) hypertrophy of goblet cells and mucus glands

46. Acute intravascular hemolysis may follow exposure to
1) naphthalene
2) stibine
3) phenylhydrazine
4) arsine

47. A sponsor of an Investigational New Drug (IND) is required to report adverse drug experience to the Food and Drug Administration within 3 working days when
1) a patient dies in a clinical study
2) an unexpected death occurs in a clinical study which was unrelated to the test drug
3) new animal toxicology data show unexpected long-term toxicity
4) an unexpected fatal or life-threatening experience associated with the use of the drug in a clinical study occurs

48. Alpha particles
1) are a type of ionizing radiation
2) lose energy when passing through biological tissue
3) are heavy charged particles
4) have linear energy transfer much greater than for gamma rays

49. Peripheral neuropathy produced by hydrocarbon solvents has which of the following characteristics?
1) dying-back neuropathy
2) may involve metabolic conversion to the ultimate neurotoxicant
3) may be potentiated by compounds such as methylethyl ketone
4) may be reversible on cessation of exposure

50. Fumonisins are secondary metabolites of *Fusarium moniliforme, F. proliferatum* and other *Fusaria* that commonly contaminate corn. These mycotoxins are reported to be associated with
1) equine leukoencephalomalacia
2) porcine edema
3) rodent hepatotoxicity
4) human esophageal cancer

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1. Which of the following always lead(s) to a nonfunctional gene product?
   A) frame shift mutation
   B) missense base pair substitution
   C) nonsense base pair substitution
   D) transition base pair substitution
   E) A and C

2. Carcinogenicity of 1,2 dichloroethane is thought to be associated with the hepatic biotransformation to
   A) a glutathione conjugate
   B) chloride ion and an epoxide
   C) carbon monoxide
   D) a glucuronide conjugate
   E) none of the above

3. Subacute myelo-opticoneuropathy was first recognized in Japan and attributed to the use of which of the following?
   A) quinine
   B) ethambutol
   C) iodochlorhydroxyquin
   D) atoxyl
   E) chlorpromazine

4. Human spina bifida, a neural tube defect, is clinically associated with which of the following drugs?
   A) thalidomide
   B) phenytoin
   C) valproic acid
   D) cocaine
   E) retinoic acid

5. Which of the following halogenated hydrocarbon anesthetics has/have been shown to produce renal failure?
   A) methoxyflurane
   B) halothane
   C) enflurane
   D) isoflurane
   E) A and B

6. Transgenic mice containing shuttle vectors carrying bacterial genes are being proposed as possible *in vivo* mutagenicity models. Which of the following is NOT a potential advantage of this model?
   A) ability to assess exposure
   B) ability to compare mutagenicity changes with the historical tumor incidence in transgenic mice
   C) ability to estimate DNA adduct levels at target tissue doses
   D) ability to evaluate the role of absorption, distribution, metabolism and excretion
   E) ability to estimate dose at target tissue

7. Which of the following chemicals is NOT an established human teratogen?
   A) aminopterin
   B) cocaine
   C) isotretinoin
   D) methyl mercury
   E) benzene

8. Which of the following, found in semen, is an indicator of seminal vesicle function?
   A) citric acid
   B) androgen-binding protein
   C) zinc
   D) fructose
   E) acid phosphatase

9. Isoniazid neurotoxicity is related to
   A) significant reduction of neural L-glutamate decarboxylase activity
   B) depletion of pyridoxine and pyridoxal from liver and peripheral nervous tissues
   C) enhanced kidney excretion of vitamin B6
   D) reduction of inhibitory transmitters in the brain
   E) all of the above

10. Allergic contact dermatitis resulting from exposure to chromium compounds is an example of
   A) anergy
   B) B-cell mediated allergy
   C) T-cell mediated allergy
   D) type I reactivity
   E) Goodpasture's disease

11. For a chemical to be considered a rodent carcinogen it must
   A) exhibit both initiating and promoting activity in an initiation/promotion model
   B) increase the incidence of tumors
   C) act through a genotoxic mechanism
   D) exhibit a short latency period
   E) all of the above

12. Potential ocular photosensitizer(s) that induce UV-cataractogenesis include
   A) dinitrophenol
   B) steroids
   C) naphthalene
   D) phenothiazines
   E) all of the above

13. True cross-sensitization reactions of the skin
   A) may occur when two or more antigens share similar groups
   B) occur only when exposure to two or more agents and sunlight occur simultaneously
   C) is seen only in allergic contact dermatitis
   D) occur only when both agents are in the same product
   E) may occur when the same antigen is present in more than one product

14. Which of the following mycotoxins is implicated in Balkan nephropathy and has a potential to induce DNA adduct formation making it a possible renal carcinogen?
   A) aflatoxin B_1
   B) aflatoxin G_1
   C) citrinin
   D) ochratoxin A
   E) rubratoxin B

15. Which of the following does not reflect a correct pairing of nephrotoxic agent with presumed site of action in the nephron?
   A) inorganic mercury - pars recta
   B) amphotericin B- proximal and distal convoluted tubule
   C) aspirin plus phenacetin - collecting duct
   D) puromycin aminonucleoside - glomerulus
   E) fluoride - loop of Henle

16. Which of the following causes a marked elevation in blood glucose when applied to abraded skin of dogs?
   A) sodium lauryl sulfate
   B) cyclic nucleotides
   C) toluene
   D) benzene
   E) polycyclic aromatic hydrocarbons

17. Which of the following is NOT characteristic of vitamin A toxicity?
   A) improper bone growth
   B) night blindness
   C) dermatitis
   D) improper *in utero* development of offspring
   E) hepatomegaly

18. Maximum percent deposition of inhaled particles will occur in this region of the respiratory tract, when an aerosol having an MMAD of 3 μm is inhaled during tidal breathing
   A) nasal
   B) oral pharyngeal
   C) peripheral lung
   D) tracheobronchial
   E) alveolar

19. Which of the following is most likely to cause both neuropathy and alopecia?
   A) lead
   B) mercury
   C) selenium
   D) beryllium
   E) thallium

20. Which of the following, when oxidized, provides reducing equivalents for ATP synthesis?
   A) carbon monoxide
   B) cyanide
   C) ethanol
   D) lead
   E) none of the above

21. Calculate the dynamic concentration of mercury vapor in an air sample produced by mixing an air stream saturated with mercury vapor (18.79 mg/m^3) with an air stream containing no mercury vapor. If the flow rates of the saturated and clean air streams are 0.04 L/min and 36.5 L/min, respectively, what is the resultant concentration of mercury vapor in air in mg/m^3?
   A) 0 .002 mg/m^3
   B) 0.02 mg/m^3
   C) 0.2 mg/m^3
   D) 2 mg/m^3
   E) none of the above

22. The neurologic syndrome that may occur as a result of alcoholic liver cirrhosis consists of
   A) ataxia, tremor and dementia
   B) dementia, polyneuritis and paresthesia
   C) psychosis, hyperactivity, chorea
   D) muscle weakness, behavior deficits and eventual paralysis
   E) C and D

23. Which statement concerning dermal sensitization is correct?
   A) is a reaction confined to the skin, and is not a systemic process
   B) a hapten penetrates the dermis and forms a union with proteins, giving rise to a complete antigen
   C) lymphocytes do not play a role in sensitization
   D) skin permeability is the same for rabbit, rat, guinea pig and humans
   E) all are correct

24. An infant shows signs that would normally be associated with an extra chromosome; however, the chromosome number is normal. Neither parent shows these signs. However, one of the mother's chromosomes is abnormally long and another is very short. One explanation for the signs is
   A) the effect is recessive and skips a generation, so the abnormality would have appeared in the grandparents instead of the parents
   B) there was a point mutation *in utero* resulting in aneuploidy
   C) there was a translocation in the maternal genome
   D) the infant was exposed to ionizing radiation *in utero*
   E) none of the above

25. After chemical exposure, appearance of sister chromatid exchanges in mammalian cells indicates
   A) point mutations
   B) histone annealing
   C) DNA intercalation
   D) DNA double strand breaks
   E) DNA deletions

26. The progenitor cells for alveolar repair are
   A) type I cells
   B) type II cells
   C) basal cells
   D) Clara cells
   E) all of the above

27. In an Ames assay, the number of revertants increases with increasing dose until a certain concentration and then frequently declines because of
   A) solubility limits
   B) cytotoxicity
   C) depletion of nutrients
   D) saturation of critical metabolic pathways
   E) oxygen depletion

28. Which of the following agents is NOT associated with the induction of an allergic response?
   A) formaldehyde Y
   B) aspirin
   C) chloroform
   D) nickel Y
   E) toluene diisocyanate

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29. Compounds that can cause occupational contact dermatitis are
   1) chromates Y
   2) lead ✓ B
   3) nickel Y
   4) borates

30. Which of the following is/are correct for interferon?
   1) is part of the host defense mechanism
   2) exists in several antigenic forms A
   3) is produced in response to viral infection
   4) is useful only in treatment of cancer

31. Methemoglobinemia can occur upon exposure to
   1) aniline Y
   2) nitrobenzene Y A ?
   3) amy nitrite Y
   4) dinitrophenol – ✓ ?

32. Bromobenzene hepatotoxicity
   1) requires metabolic activation to a reactive epoxide intermediate T
   2) phenobarbital, SKF-525A and piperonyl butoxide enhance toxicity T A ?
   3) is increased by diethyl maleate administration
   4) requires mixed function oxidase activity for detoxification ✓ Is epoxide hydrolase or MFO

33. Which of the following cause interstitial pulmonary fibrosis?
   1) busulfan Y
   2) methotrexate Y B
   3) paraquat Y
   4) prednisolone

34. Acute treatment of rats with a compound which selectively destroys spermatogonial cells
   1) may produce immediate and lasting sterility C
   2) may cause irreversible sterility
   3) will always be reversible
   4) will cause sterility after about 10 weeks post-dosing

35. Which of the following require(s) metabolism or bioactivation to be teratogenic?
1) acetylaminofluorene
2) busulfan
3) thalidomide
4) nitrogen mustard

36. Which of the following statements is/are correct?
1) permeability of the blood-testis barrier to non-electrolytes is dependent on molecular size
2) permeability of the blood-testis barrier to ionic compounds does not correlate with lipid solubility
3) no blood-ovary barrier exists to protect the ovary from toxic chemicals carried in the blood
4) the male germ cells have no effective barrier to toxic chemical agents carried in the blood

37. Which of the following can be classified as an autoimmune disease?
1) systemic lupus erythematosis
2) rheumatoid arthritis
3) Goodpasture's disease
4) serum sickness

38. Haber's rule provides a simple and practical means of assessing acute inhalation toxicity by combining which of the following measures?
1) temperature
2) exposure time
3) chamber volume
4) airborne concentration

39. Macrophages
1) have important roles in both humoral and cell-mediated immune responses
2) develop from monocytes which migrate into tissues from blood vessels
3) process and present antigens to lymphocytes
4) are not strongly influenced by lymphokines

40. Treatment of psoriasis with 8-methoxypsoralen plus whole body UV-A radiation involves the risk of cataract formation because of
1) intercalation of 8-methoxypsoralen with DNA in lens
2) accumulation of UV-A derived protein photoproducts in lens
3) UV-A induced DNA cross-linking in lens
4) UV-A induced hyperpigmentation of lens

41. Kidney damage associated with acute chloroform exposure includes which of the following?
1) increased kidney weight
2) swelling of the tubular epithelium
3) marked necrosis of the proximal tubular epithelium
4) glomerular damage

42. Ocular tissues which are routinely evaluated during Draize eye irritation testing include
1) cornea
2) lens
3) iris
4) retina

43. A new chemical has been shown to depress all aspects of cell-mediated immunity. Which of the following parameters would DECREASE following exposure?
1) proliferative response of lymphocytes to allogenic leukocytes
2) susceptibility to influenza challenge
3) delayed-type hypersensitivity response to a contact sensitizer
4) B-cell progenitor cell numbers

44. Percutaneous absorption
1) involves active transport of chemicals across the highly hydrated stratum granulosum
2) involves passive diffusion across the dried, keratin-filled cells of the stratum corneum
3) occurs predominantly via passive diffusion across the hair follicles, sweat ducts and sebaceous glands
4) varies depending on which part of the body is being examined

45. Aflatoxin-induced hepatocarcinogenesis is
1) associated with the apparent carcinogen-selective mutation in p53
2) highly correlated with the extent of total DNA adducts formed *in vivo*
3) greater in people who are hepatitis B-positive
4) due to *in vivo* formation of a metabolite, aflatoxicol

46. α_2-microglobulin nephropathy is caused by exposure to a diverse group of compounds including d-limonene and lindane. Which of the following is/are correct?
1) it is a species-specific but not a sex-specific lesion
2) droplets of low molecular weight protein accumulate in proximal tubular epithelium
3) humans are at considerable risk from α_2-microglobulin induced nephropathy
4) it may progress to the formation of renal neoplasms by a nongenotoxic mechanism

47. Classic penicillin allergy involves
1) immune recognition of the penicillin molecule
2) metabolism of penicillin to form a hapten-protein complex
3) complement-mediated cytotoxicity
4) IgE-mediated mast cell degranulation

48. Transgenic mutation assays in rodents generally employ an inserted bacterial gene such as the lacI or lacZ genes from the beta-galactosidase operon. The rationale for this approach is that

1) lacI and lacZ genes are important oncogenes and are therefore relevant to human carcinogenesis
2) lacI and lacZ genes are well characterized genes that are useful for studying mutagenesis C
3) lacI and lacZ genes are the only genes that have successfully been inserted into the rodent genome
4) it permits measurement of mutations from any rodent tissue from which DNA can be isolated

49. Which of the following agents is/are considered to be genotoxic carcinogen(s) in rodents?

1) 2-acetylaminofluorene
2) diethylnitrosamine
3) 7,12-dimethylbenz[a]anthracene A
4) 2,3,7,8-tetrachlorodibenzo-p-dioxin (TCDD) AH Receptor

50. Natural killer cells

1) are large granular lymphocytes A
2) are able to lyse certain tumor cells
3) do not require sensitization for host defense
4) ~~are phagocytic~~

ABT RECERTIFICATION EXAMINATION #17

May 12, 2000 ***PART C***

This examination constitutes part of the recertification process for toxicologists certified by the American Board of Toxicology in 1991 and 1996. There are three parts in this exam and an answer sheet is provided for each part. These answer sheets must be returned to ABT, P.O. Box 30054, Raleigh, NC 27622 for grading. An analysis of your performance in each part of the exam will be provided to the recertification committee of ABT and they will contact you regarding this and other requirements of the recertification process. If you have any questions please call the ABT Office at 919-847-8720.

EXAMINATION CATEGORIES

1. Metals, solvents, pesticides, gases, dusts and aerosols, industrial chemicals and radiation, plant and animal toxins, food additives, drugs and cosmetics.
2. Mutagenesis, carcinogenesis, developmental and reproductive toxicology, pulmonary toxicology, neurologic toxicology, immunologic toxicology, cutaneous toxicology, ocular toxicology, hepatic toxicology, hematopoietic toxicology, renal toxicology.
3. General principles, toxicokinetics, toxicodynamics and mechanisms of action, risk assessment, epidemiology and biostatistics, regulatory toxicology and risk management, environmental toxicology, occupational toxicology, forensic and clinical toxicology.

INSTRUCTIONS

1. Use a soft (#2) pencil to fill the box corresponding to your answer.
2. Select only one answer for each question. (Erase others)
3. Do not leave any questions unanswered. (No penalty for guessing).
4. Sign answer sheet ONLY on line that says name.
5. Do not bend answer sheets.

Good luck!

Key for the following questions
Choose the single most appropriate answer

1. Which of the following types of epidemiological studies would be most useful in pinpointing industrial plants or communities that have an elevated rate of a particular disease?
   A) prospective study
   B) retrospective study
   C) case-control study
   D) cross-sectional study
   E) none of the above

2. If a 12 kg dog is given a diet containing 800 ppm of a compound and consumes 400 g/day, what is the dose relative to body weight?
   A) 54 mg/kg/day
   B) 27 mg/kg/day
   C) 14 mg/kg/day
   D) 5.4 mg/kg/day
   E) none of the above

3. How many half-lives does it take for a continuously infused xenobiotic to reach 95% of a plateau plasma concentration?
   A) 1
   B) 2
   C) 5
   D) 10
   E) >10

4. A new product that claims sunscreen protection and skin wrinkling prevention is about to be marketed and contains previously approved ingredients. Which of the following is true regarding the marketing of this product in the U.S.?
   A) it is a cosmetic under the Food, Drug and Cosmetic Act, and no premarket approval is required
   B) the product is an over-the-counter drug and is covered under the Food and Drug Administration monograph for sunscreens
   C) the product is a new drug and must be developed under the Food and Drug Administration's IND/NDA system
   D) an Environmental Protection Agency premanufacturing notification is required
   E) the Consumer Product Safety Commission must be notified prior to marketing the product

5. Nickel carbonyl, a volatile liquid often associated with electroplating, nickel refining and electronic industry operations, produces lung damage by
   A) undergoing *in situ* hydrolysis in the peripheral regions of the lung and producing nascent hydrogen chloride
   B) reacting with cellular constituents to release carbon monoxide, which in turn interferes with gas exchange
   C) decomposing to metallic nickel in the periphery of the lung, with resultant cellular damage
   D) direct stimulation of beta adrenergic receptors in the lung
   E) causing type I cell hyperplasia

6. Which of the following laws authorizes the responsible U.S. agency to prescribe the required toxicity testing of appropriate compounds but does not give the agency premarketing approval responsibility?
   A) 1938 Food, Drug Cosmetic Act
   B) 1958 Food Additives Amendment
   C) 1970 Federal Insecticide, Fungicide and Rodenticide Act
   D) 1976 Toxic Substances Control Act
   E) 1976 Resource Conservation and Recovery Act

7. Which of the following parameters best reflects the development of a disease over a defined period of time?
   A) incidence rate
   B) prevalence rate
   C) mortality rate
   D) age-specific mortality rate
   E) case fatality rate

8. An intravenous infusion of 100 µg/kg/min of Drug X into a dog results in a steady-state plasma concentration of 100 ng/mL achieved 30 min after the infusion starts. Assume no saturation of excretion of metabiolism. If the rate of infusion was 200 µg/kg/min, the steady-state plasma concentration
   A) would be reached in 15 min and would be 200 ng/mL
   B) would be reached in 30 min and would be 200 ng/mL
   C) would be reached in 15 min and would be 100 ng/mL
   D) would be reached in 60 min and would be 200 ng/mL
   E) cannot be predicted from the information given

9. A volatile chlorinated hydrocarbon has a blood:air partition coefficient of 100. At low inhaled concentrations, the blood concentration at steady-state is only twice the inhaled air concentration. The most likely explanation is
   A) the compound is an irritant so the animals reduce their breathing rate
   B) chlorinated hydrocarbons do not pass the capillary membranes in the lung easily and are not equilibrated between air and blood during inhalation exposures
   C) there is extensive systemic metabolism in the animal
   D) partition coefficients are only valid for studies done *in vitro* and do not predict *in vivo* uptake
   E) the volatility is not sufficient to allow concentration of the chemical by a factor of 100 in the animal blood

10. Which of the following metabolic reactions is conducted by the gastrointestinal microflora?
   A) glucuronidation
   B) aliphatic hydroxylation
   C) arysulfaction
   D) nitroreduction
   E) epoxidation

11. If you were interested in identifying whether a chemical exposure resulted in the increase in transcription of a particular gene, which of the following tests would be most useful?
   A) southern blot analysis DNA
   B) western blot analysis ?
   C) northern blot analysis RNA
   D) DNA footprinting
   E) ribonuclease protection assay

12. All of the following are examples of xenobiotics activated by human cytochrome P450 2E1 EXCEPT
   A) acrylonitrile
   B) styrene
   C) benzene
   D) benzo(a)pyrene
   E) chloroform

13. Attributable risk can be defined as
   A) the proportional excess risk of disease that is associated with exposure to a risk factor
   B) ~~the risk of disease in the total population~~
   C) the relative risk of disease in two different populations
   D) the odds ~~ratio for disease in the~~ exposed population
   E) the incidence ~~of disease in the exposed group~~

14. Relative risk is
   A) the ratio of incidence rate of disease in the exposed group to incidence rate in the total population
   B) the ratio of incidence rate of disease in the exposed group to incidence rate in the ~~unexposed group~~
   C) identical to the odds ratio
   D) the incidence of disease in the exposed group
   E) the proportional excess risk of disease that is associated with exposure to a risk factor

15. Glycol ethers were at one time considered to be very safe solvents for use in coating formulations. Their use has been greatly curtailed due to concerns regarding adverse effects on the
   A) reproductive system
   B) urinary system
   C) skin
   D) lungs
   E) liver

16. Percutaneous penetration by chemicals is, in general, highest in which anatomical region(s) of the human body?
   A) forearm
   B) palm of hand and instep of foot
   C) abdomen
   D) trunk
   E) head and scrotum

17. Bladder tumors have been associated with occupational exposure to
   A) chloroform
   B) vinyl chloride Hemangiosarcoma.
   C) benzene
   D) beta-naphthylamine
   E) vinyl acetate

18. A cause of severe respiratory symptoms as a result of welding is
   A) ozone
   B) sulfur dioxide
   C) acrolein
   D) cadmium fumes
   E) silver fumes

19. Exposure to cold temperatures potentiates the toxicity of
   A) caffeine
   B) digitalis
   C) ethanol
   D) nicotine
   E) parathion

Malathion, Atropine, Nicotine, Strychnine

20. Which is NOT a suitable means for measuring biodegradability?
- A) absorption of infrared radiation
- B) biochemical oxygen demand relative to theoretical oxygen demand
- C) disappearance of parent compound
- D) evolution of carbon dioxide
- E) loss of dissolved organic carbon

21. In a conscious patient, gastric lavage is contraindicated in the treatment of poisoning by which of the following agents?
- A) barbiturates
- B) phosphorus
- C) iron
- D) amphetamine
- E) sodium hypochlorite (bleach)

22. Which combination is an example of antagonism due to competition for binding at the target site?
- A) aldrin and tri-ortho-cresol phosphate
- B) TOCP and sarin
- C) sarin and EPN (O-ethyl O-(4- nitrophenyl) phenylphosphonothioate
- D) physostigmine and sarin
- E) none of the above

23. Which of the following toxicants produces effects on the hematopoietic system resembling the effects of radiation exposure?
- A) nerve gases
- B) beryllium
- C) fluoroacetate
- D) nitrogen mustards
- E) chlorinated hydrocarbons

24. At a dose of radiation to the whole body of 5 Gray (500 rad), mammals, including man, will die in
- A) 1 - 2 days
- B) 3 - 5 days
- C) 6 - 10 days
- D) 1 - 2 months
- E) no effect on life - span

25. Excess risk = q*d where q* is the
- A) 95% upper confidence limit on risk per unit dose
- B) normally distributed population of a quantal dose-response curve
- C) percent of positive cancers found in a lifetime study
- D) logarithm of potency of the test substance
- E) lifetime average daily dose expressed as mg/kg/day

26. According to U.S. FDA GLP regulations, the study director should be
- A) an experienced, board-certified toxicologist
- B) scientist or other professional of appropriate education, training and experience
- C) a skilled manager with authority to alter study design
- D) a scientist reporting to the quality assurance unit
- E) an employee of the testing facility with a Ph.D. or equivalent

27. Of the following occupationally encountered compounds, which presents a greater dermal than inhalation exposure hazard?
- A) dimethylformamide
- B) carbon oxysulfide
- C) elemental mercury
- D) alpha aminolevulinic acid
- E) n-isopropylaniline

28. In a normally distributed population, +/- 1 SD about the mean represents approximately what percentage of the population?
- A) 10%
- B) 20%
- C) 50%
- D) 70%
- E) 90%

29. Under the U.S. Toxic Substances Control Act, the EPA requires inclusion in the reports on new chemicals all of the following EXCEPT
- A) total amount manufactured
- B) description of byproducts resulting from the use of the chemical
- C) changes in the formulation of mixtures if no significant effects are expected
- D) occupational exposures to the chemical including number of people exposed
- E) methods of disposal of the chemicals

30. Which of the following chemicals has/have been suggested to produce liver tumors in laboratory animals as a result of chronic cytotoxicity?
- A) tetrachloroethylene
- B) phenobarbital
- C) 2-biphenylamine
- D) all of the above
- E) none of the above

31. For a subacute study in rats weighing 250 gm, the test agent is formulated as a 5% suspension in a 0.3% corn starch vehicle. To give daily doses of 50, 150 and 500 mg/kg, the toxicologist should employ dose volumes of
A) 1, 3, and 10 ml/kg for the three treatment groups
B) 16.7, 50, and 167 ml/kg for the three treatment groups
C) 0.1, 0.3, and 1.0 ml/kg for the three treatment groups
D) 10, 30, and 100 ml/kg for the three treatment groups
E) 100, 300 and 1000 ml/kg for the three treatment groups

32. After intravenous injection, cefazolin has a $t_{1/2}$ of 2 hr in humans, 0.5 hr in dogs and 0.3 hr in rabbits. Ceforanide has a $t_{1/2}$ of 3 hr in humans, 1.2 hr in dogs and 5-9 hr in rabbits. Both cephalosporins were nephrotoxic in the rabbit but not in humans, dogs, or monkeys. Choose the statement that represents the best conclusion from this information.
A) the rabbit represents a good model for safety evaluation studies of cefazolin and ceforanide
B) nephrotoxicity is not related to the pharmacokinetics of cephalosporins
C) a comparative acute nephrotoxicity study on the two cephalosporins in the rabbit may not be valid for extrapolation to the human
D) cefazolin is safer for human use than ceforanide
E) neither compound should be recommended for human use

Key for the following questions
Choose A) if 1,2 and 3 are correct
Choose B) if 1 and 3 are correct
Choose C) if 2 and 4 are correct
Choose D) if only 4 is correct
Choose E) if all four answers are correct

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33. The U.S. Animal Welfare Act has specific requirements for humane handling of
1) rabbits
2) dogs
3) nonhuman primates
4) rats

34. Admissions to a large hospital for myocardial infarction (MI) over a period of one year were analyzed for effects of the use of oral contraceptives (OC) to determine if there was any relationship. Of 100 women with MI, 70 were using OC. Of 100 women admitted with problems other than MI, 30 were using OC.
1) the relative risk (RR) can be calculated directly and is 2.3
2) the odds ratio (OR) provides an estimate of relative risk and is 5.4
3) if the study groups were larger the RR and OR would be closer together
4) the study is a case-control study

35. Which of the following is/are example(s) of mechanistic extrapolation models?
1) linearized multistage
2) log-probit
3) gamma multihit
4) Weibull

36. Which of the following is/are correct concerning the concepts of strength of evidence and weight of evidence?
1) strength of evidence refers to the degree of conviction regarding the outcome of an experiment ("clear evidence of carcinogenic activity")
2) strength of evidence refers to carcinogenic potency or to mechanism of action of a chemical
3) weight of evidence involves consideration and integration of all relevant human and long-term animal studies
4) weight of evidence refers to the amount and kind of data from a single experiment

37. Which of the following statements regarding the concept of statistical power is/are true?
1) the power of a study refers to the probability that a statistical test of the results will identify a positive effect when one exists
2) the power of a study is the probability that it will lead to the correct rejection of the null hypothesis
3) the significance level, alpha is the probability of committing a Type I error, i.e, rejecting a true null hypothesis
4) to compute the power of a study, the investigator must specify all of the following parameters: alpha, beta, effect size, sample size and variance

38. The polymerase chain reaction has been used for
1) messenger RNA quantitation
2) analysis of point mutations
3) amplifying ancient DNA
4) analyzing DNA from paraffin-embedded tissues

39. Under Good Laboratory Practice regulations, which of the following is/are NOT functions of the Quality Assurance Unit?
1) maintenance of a copy of the master schedule
2) maintenance of copies of all applicable protocols
3) inspection of each study to ensure its integrity
4) overall responsibility for the technical conduct of the study

40. Which of the following statistical methods is/are parametric procedures?
1) analysis of variance
2) Wilcoxon rank sum test
3) Student's t-test
4) Kruskal-Wallis test

41. Which of the following statements(s) is/are true about the standard error of the mean (SEM)?
1) the standard deviation (SD) is considered to be a better indicator of variability in individual measurements than the SEM
2) the SEM depends on both the SD of the original population and the size of the sample
3) the SEM is used to state the uncertainty of the estimate of the mean derived from a sample of the population
4) the SEM is a direct estimate of the variability of the data from which the mean is calculated

42. Components of a subchronic neurotoxicity evaluation in the rat under U.S. FIFRA guidelines include
1) functional observational battery (FOB)
2) motor activity monitored by an automated activity recording apparatus
3) histopathological evaluation of the central and peripheral nervous system
4) measurement of neurotoxic esterase activity (NTE)

43. Which of the following elements has/have a specific binding protein associated with its absorption or distribution?
1) Cu
2) Fe
3) Ca
4) Zn

44. Which of the following is/are true with respect to the reference dose (RfD)?
1) can be defined as an estimate of a daily exposure level for an individual that is not likely to cause a deleterious effect
2) is frequently derived by dividing the NOAEL by appropriate uncertainty and modifying factors
3) can be established from a benchmark dose using appropriate uncertainty and modifying factors
4) are protective for systemic as well as carcinogenic effects

45. Which of the following statements is/are correct?
1) in an epidemiological evaluation of carcinogenicity, the observed/expected ratios of cancer deaths are: lung 4.79, stomach 1.63, colon 1.59, kidney 0.82. Therefore the exposure is most likely associated with kidney cancer
2) in the absence of tobacco smoking, the asbestos worker does not appear to be at increased risk of developing mesothelioma
3) all persons occupationally exposed to an amount of asbestos sufficient to cause mesothelioma will show lung X-ray changes typical of the asbestosis
4) pathogenicity and carcinogenicity of asbestos fibers are a function of their type and length

46. Gastric lavage
1) is often preferred to induced emesis
2) usually employs tap water
3) is less likely to produce aspiration pneumonia than is emesis
4) is useful in comatose patients

47. The U.S. Food, Drug and Cosmetic Act passed in 1938 provides for
1) extension of legislation from food alone to cosmetics and medical devices
2) establishment of tolerances for unavoidable or required poisonous substances
3) authorized factory inspections
4) predistribution clearance on safety of new drugs

48. Which of the following classes of toxicants is/are usually presumed to have no threshold dose in assessing risk?
1) delayed neurotoxicants
2) mutagens
3) teratogens
4) carcinogens

49. Which of the following produce(s) both acute and chronic toxic effects?
1) carbon monoxide
2) silicon dioxide → Asbestosis agent
3) arsenic trioxide
4) hypoxemia

50. Microbial metabolism is of toxicologic significance in
1) methylation of inorganic mercury to mono- or di-methylmercury
2) methylation of inorganic arsenic to trimethyl arsine
3) N-nitrosation of secondary and tertiary amines and/or quaternary nitrogen compounds
4) conversion of trimethyltin hydroxide to phenyltin → immunosuppressant